DANISH HANIF MIDTERM FILES
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MTH001 Quiz File

السلام وعلیکم!

امید کرتا ہوں کہ آپ سب خیریت سے ہوں گے کافی دنوں کی محنت کے بعد الحمدللہ ورچوئل یونیورسٹی کے مڈٹرم امتحان کے لیے فائلز تیار کرلی ہیں امید کرتا ہوں کے آپ سب سٹوڈنٹس کے امتحانات میں بہت فائدہ مند ہوں گی۔ فائلز میں کسی قسم کی کوئی بھی غلطی ہو گئی ہو تو اس کے لیے معذرت خواہ ہیں انشاءاللہ مذید محنت سے ان فائلز کو بہتر بنائیں گے۔ آپ سب سٹوڈنٹس سے گزارش ہے کہ ہمارے یوٹیوب چینل اور ویبسائیٹ کو فالو کرلیں۔

مکمل پیپرز آپکو میری ویبسائیٹ پہ ملیں گے لنک نیچے دیا گیا ہے۔

Here is Website Link: https://hbdanish.me/

Here is YouTube Channel Link:

https://www.youtube.com/@Danish_Khagga

اگر آپ لوگ اپنا ایل۔ایم۔ایس ہینڈل کروانا چاہتے ہیں یا اسائنمنٹس کوئز کروانا چاہتے ہیں تو اس نمبر پہ واٹس ایپ کریں۔

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Question #1:

If $U = \{1, 2, 3, 4, 5, 6, 7\}, A = \{1, 3, 5, 7\}, B = \{1, 2, 3, 4, 6\}$, then $A \cup B$ is:

a) $\{1, 2, 3, 4, 5, 6, 7\}$

b) $\{1, 3\}$

c) $\{1\}$

d) $\{\}$

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Answer: a) ({1,2,3,4,5,6,7}

Question #2:

Let p and q be two statements, then the conjunction of p and q is denoted as:

a) $p \rightarrow q$

b) $p \wedge q$

c) $p \vee q$

d) $p \leftrightarrow q$

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Answer: b) $p \wedge q$

Question #3:

Let $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2), (2, 3), (3, 3), (3, 4)\}$ be a relation on A.
Which one of the following ordered pairs has made R an irreflexive relation?

a) $(1, 2)$

b) $(3, 3)$

c) $(3, 4)$

d) $(2, 3)$

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Answer: b) $(3, 3)$

$\{x \mid x$ belongs to the set of natural numbers and $2 < x < 4\}$ is:

a) Empty set

b) A finite set

c) An infinite set

d) A singleton set

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Answer: a) Empty set

Consider that q and p are statements such that $q \rightarrow p$ is false. Then the truth value of $q \leftrightarrow p$ is:

a) True

b) False

Answer: b) False

The range of the relation $\{(0, 1), (3, 22), (90, 34)\}$ is:

a) $\{0, 1, 3\}$

b) $\{0, 3, 90\}$

c) $\{0, 1, 3, 22, 90, 34\}$

d) $\{1, 22, 34\}$

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Answer: d) $\{1, 22, 34\}$

Question #7:

If A is a subset of the universal set U, then $A \cap \overline{A}$ is:

a) A

b) Empty set

c) U

d) Complement of A

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Answer: b) Empty set

Consider that q and p are statements such that $q \rightarrow p$ is false. Then the truth value of $q \rightarrow \sim p$ is:

a) False

b) True

Answer: a) False

If $U = \{a, b, c, d, e, f, g\}, A = \{a, c, e, g\}$, and $B = \{d, e, f, g\}$, then $A \cap B$ is:

a) $\{e, c\}$

b) $\{a, c, d, f, g\}$

c) $\{\}$

d) $\{e, g\}$

Answer: d) $\{e, g\}$

If in a directed graph of a relation R on A there is no loop at each point of A, then R is:

a) An irreflexive relation

b) Not an irreflexive relation

c) Not a transitive relation

d) A transitive relation

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Answer: a) An irreflexive relation

Question #1: Let A = {1, 2, 3} and R = {(1,2), (2,3), (1,3), (2,1)} be a relation on A. Which of the following ordered pairs are needed in R to make it transitive?

A) (1,1) and (2,2)
B) (2,2) and (3,2)
C) (3,1) and (3,2)
D) (1,1) and (3,3)

Correct Answer: D) (1,1) and (3,3)

Question #3: If A=$\{3,5,6,9,10\}$ $A=\{3,5,6,9,10\}$ and B=$\{2,7,8,12\}$ $B=\{2,7,8,12\}$, then $A\cap B$ equals:

Select the correct option:

A) $\{2,3,5,6,7,8,9,10,11,12\}$ $\{2,3,5,6,7,8,9,10,11,12\}$

B) $\{2,3,5,6,7,8,9,10,12\}$ $\{2,3,5,6,7,8,9,10,12\}$

C) Empty set

D) $\{6\}$ $\{6\}$

Correct Answer: B) $\{2,3,5,6,7,8,9,10,12\}$ $\{2,3,5,6,7,8,9,10,12\}$

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Question #4: In $p\rightarrow q$, the statement q is called _____.

Select the correct option:

A) Antecedent

B) Hypothesis

C) Conditional

D) Conclusion

Correct Answer: D) Conclusion

Question #5: If N=$\{-5,-4,0,6,8\}$ $N=\{-5,-4,0,6,8\}$ and O=$\{-4,0,8,9\}$ $O=\{-4,0,8,9\}$, then $N\cup O$ equals:

Select the correct option:

A) $\{-5,-4,0,6,8,9\}$ $\{-5,-4,0,6,8,9\}$

B) $\{0,8\}\{0,8\}$

C) $\{-4,0,8\}\{-4,0,8\}$

D) $\{-4,0\}\{-4,0\}$

Correct Answer: A) $\{-5,-4,0,6,8,9\}\{-5,-4,0,6,8,9\}$

Question #6: Let $A=\{0,1,2\}$ and $R=\{(0,2),(1,1),(2,0)\}$ be a relation on A. Then R is _____.

Select the correct option:

A) Reflexive

B) Transitive

C) Symmetric

D) None of the above

Correct Answer: C) Symmetric

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Question #7: If A is a subset of the universal set U, then $\bar{A}\cup A$ equals:

Select the correct option:

A) A

B) Empty set

C) Complement of A

D) U

Correct Answer: D) U

Question #9: The statement form $\vee\sim p\vee\sim p$ is _____.

Select the correct option:

A) Falsity

B) Negation of q

C) Tautology

D) Negation of p

Correct Answer: C) Tautology

Question #10: Consider that q and p are statements such that $q\rightarrow p$ is false. Then the truth value of $q\rightarrow\sim p$ is _____.

Select the correct option:

A) True

B) False

Correct Answer: A) True

Question #1: If the "common ratio = 3" for the following geometric sequence: (x-1), (x-1)(x-1), (x-1)(x-1)(x-1), ---------, then the value of 'x' is:

A) 4

B) 2

C) 1

D) 3

Correct Answer: C) 1

Question #2: The 9999th term of the sequence: {-1, 1, -1, 1, -1, 1, ...} is:

A) 9999

B) 1

C) -1

D) -9999

Correct Answer: C) -1

Question #3: How many one-to-one functions are there from a set with three elements to a set with four elements?

A) 4 + 3 × 2 = 10

B) 3 × 4 × 1 = 12

C) 4 × 3 + 2 = 14

D) 4 × 3 × 2 = 24

Correct Answer: B) 3 × 4 × 1 = 12

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Question #4: If the sum of the first n natural numbers is n(n+1)/2, then their average will be:

A) (n+1)/2

B) n/2

C) n(n+1)/2

D) (n-1)/2

Correct Answer: A) (n+1)/2

Question #5: A graph of a function f is one-to-one iff every horizontal line intersects the graph in at most _____.

A) one point

B) two points

Correct Answer: A) one point

Question #6: The graph of y = square of x is a straight line.

A) False

B) True

Correct Answer: A) False

Question #7: Sum of the first 2000 terms of the sequence: {-1, 1, -1, 1, ...} is

A) Undetermined

B) 0

C) 1

D) -1

Correct Answer: B) 0

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Question #8: A constant function is one-to-one if and only if

A) Its co-domain is empty

B) Its domain is empty

C) Its domain is singleton

D) Its co-domain is singleton

Correct Answer: C) Its domain is singleton

Question #9: For an arithmetic sequence: $x + a^2$, $x + 3a^2$, $x + 5a^2$, …. if the common difference is '4' then the value of 'a' is -------.

A) ±1

B) ±Square root of '2'

C) 0

D) ±2

Correct Answer: B) ±Square root of '2'

Question #10: The function f(x) = x + 1 from the set of integers to itself is onto.

A) True

B) False

Correct Answer: B) False

Question #1: Geometrical criterion for the graph to be a function is that any---------in its domain intersects the graph in at most one point.

a) Vertical line

b) Line at angle 60 degrees

c) Line at angle 45 degrees

d) Horizontal line

Answer: a) Vertical line

Question #2: Identify the sequence as Arithmetic, Geometric, or neither. 2, 6, 8, 54, 62, . . .

a) Harmonic

b) Arithmetic

c) Geometric

d) Neither

Answer: c) Geometric

Question #3: The explicit formula for the sequence 15, 12, 9, 6, . . . is _____.

a) $a_n=15-(n-1)\times 3$

b) $a_n=15\times(n-1)\times 3$

c) $a_n=15+(n-1)\times 3$

d) $a_n=15-(n-1)\times 3$

Answer: d) $a_n=15-(n-1)\times 3$

Question #4: Is the set $\theta=\{(X,|X|):X\subseteq Z_5\}$ a function? If so, what is its range?

a) $\{0,1,2,3,4,5\}\{0,1,2,3,4,5\}$

b) $\{1,2,3,4\}\{1,2,3,4\}$

Answer: b) $\{1,2,3,4\}\{1,2,3,4\}$

Question #5: If f is a function from $A=\{a,b,c\}$ to $B=\{1,2,3\}$ such that $f=\{(a,3),(b,1),(c,2)\}$, then f is called _____.

a) None of these

b) Injective function

c) Surjective function

d) Bijective function

Answer: d) Bijective function

Question #6: If $f:\mathrm{R}\to\mathrm{R}$ and $f(x)=x1$, then "_____" is true for the function is not onto.

a) There is no x in Y

b) There is no x in R

c) There is no x in R that has output = 0

d) The function is onto.

Answer: c) There is no x in R that has output = 0

Question #7: Consider the cosine function $\cos:\mathrm{R}\to\mathrm{R}$. Is this function injective?

a) False b) True

Answer: b) True

Question #8: If f is a function from A to B, then the set of elements of B is called _____.

a) None of these

b) Domain

c) Co-domain

d) Range

Answer: c) Co-domain

Question #9: If f is a function from $A=\{1,2\}$ to $B=\{a,b,c\}$ such that $f=\{(1,a),(2,b)\}$, then f is called _____.

a) Bijective function

b) None of these

c) Injective function

d) Surjective function

Answer: c) Injective function

Question #10: The function $f(x)=x+1$ from the set of integers to itself is onto.

a) True

b) False

Answer: a) True

Question 3 of 10:

Consider the arithmetic series with the following parameters:

$a_1=-3, d=2, S_n=21$

What is the number of terms (n) in this arithmetic series?

A) 10

B) 5

C) -7

D) 7

Answer: B) 5

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Question 4 of 10:

How many functions are there from a set with two elements to a set with three elements?

A) 2

B) 3

C) 6

D) 9

Answer: C) 6

Question 5 of 10:

The explicit formula for an arithmetic sequence when $a_1=6$ and $d=2$ is _____.

A) $a_n=6+2(n-1)$

B) $a_n=6+2(n+1)$

C) $a_n=6+2n$

D) $a_n=6-2(n-1)$

Answer: C) $a_n=6+2n$

Question 6 of 10:

Identify the sequence as Arithmetic, Geometric, or neither:

$8, 13, 18, 23, \ldots$ $8,13,18,23,\ldots$

A) Harmonic

B) Neither

C) Arithmetic

D) Geometric

Answer: C) Arithmetic

Question 7 of 10:

The range of the function is always a _____ of the co-domain of the function.

A) Null Set

B) Proper Subset

C) Subset

D) Power set

Answer: C) Subset

Question 8 of 10:

_____ is called a bijective function.

A) (1-1) and onto function

B) Into function

C) Onto function

D) None of these

Answer: A) (1-1) and onto function

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Question 9 of 10:

Let $f:\mathrm{R}\to\mathrm{R}$ be defined by the rule $f(x)=2x+1$ for all $x\in\mathrm{R}$. The function is_____.

A) Identity

B) Not onto

C) Not one-to-one

D) One-to-one

Answer: D) One-to-one

Question 10 of 10:

The images of all non-zero positive real numbers for the function $f(x)=-x$ are _____.

A) Never real numbers

B) Also real numbers

Answer: A) Never real numbers

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